

- 5 (a) (i) The gravitational field strength g at a point is defined as the gravitational force per unit mass acting at that point.

**A common mistake made is field strength was defined in terms of force on a unit mass rather than force per unit mass.*

- (ii) Newton's law of gravitation states that the force of attraction between two point masses is directly proportional to the product of the masses and inversely proportional to the square of the distance between them.

From this definition, considering two point masses M and m separated a distance R apart, the gravitational force that M exerts on m is $F = \frac{GMm}{R^2}$. Since gravitation

field strength is the force per unit mass m , $g = \frac{F}{m} = \frac{GMm}{R^2} \frac{1}{m} = \frac{GM}{R^2}$

**In the definition, the masses must be stated as point masses.*

(b) (i) $\rho = \frac{M}{V} = \frac{5.2 \times 10^{30}}{\frac{4}{3}\pi(1.7 \times 10^4)^3} = 2.5 \times 10^{17} \text{ kg m}^{-3}$

- (ii) The density is likely to be larger nearer the centre. The closer it is to the centre, the greater the amount of matter above with gravity pressing inward and exerting a greater pressure, forcing the matter to be more closely packed. Since gravitational force varies inversely with the square of separation, the closer spacing between the matter near the centre will pull them even closer, making the core denser.

**Very few quoted that the outer layers cause compression on the inner layers. The most common answer was to assume constant mass and then to state that volume depends on radius and hence increasing the radius will lower the density.*

$$GM = (6.67 \times 10^{-11})(5.2 \times 10^{30})$$

$$\text{(ii)} \quad a = R\omega^2 = R\left(\frac{2\pi}{T}\right)^2 = (1.7 \times 10^4) \left(\frac{2\pi}{0.21}\right)^2 = 1.5 \times 10^7 \text{ m s}^{-2}$$

- (ii)** No. The particles will not leave the surface. The gravitational field strength ($1.2 \times 10^{12} \text{ N kg}^{-1}$) is much greater than the centripetal acceleration ($1.5 \times 10^7 \text{ m s}^{-2}$). Therefore, the gravitational force acting on a particle on the surface is more than sufficient to provide the centripetal force required to keep it in circular motion about the centre of the star. This will also mean that there will be a non-zero normal contact force acting on the particle by the star, causing particle to remain in contact with the surface of the star.

**Many just merely stated that the particles would not fly off because the field strength is greater than the centripetal acceleration without any further elaboration.*

- (d)** The strong gravitational field strength accelerates the protons to very high speeds as they stream towards the star. As the protons collide with the star, they are decelerated and loses kinetic energy, and this loss of energy is emitted as a electromagnetic radiation. The wavelength of the radiation is inversely proportional to the energy of the radiation. Since protons have large kinetic energy, the energy of the radiation emitted will be large, resulting in the radiation to have a short wavelength and can form the X-ray spectrum.

- 6 (a)** The gradient of the graph is not constant, indicating the acceleration of the ball